

GLENWOOD, AB, Canada

WMO: 718850

Lat: **49.334N**Lon: **113.526W**Elev: **1074**StdP: **89.07**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.0	-24.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	30.1	N/A	28.2	N/A	26.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-4.9	-6.3	-1.2	4.0	10.5	13.7	16.7	16.8	12.5	5.3	0.0	-6.4	(6)	
(7)		DBStd	10.41	8.08	8.64	7.22	5.52	4.26	3.30	3.01	3.46	4.88	5.81	6.99	8.32	(7)	
(8)		HDD10.0	2510	461	457	346	188	47	6	0	2	29	163	301	510	(8)	
(9)		HDD18.3	4871	719	691	604	428	245	144	68	71	180	403	548	768	(9)	
(10)		CDD10.0	731	0	0	0	9	61	116	207	213	105	18	2	0	(10)	
(11)		CDD18.3	52	0	0	0	0	1	4	17	23	6	0	0	0	(11)	
(12)		CDH23.3	1278	0	0	0	7	54	97	416	504	193	7	0	0	(12)	
(13)		CDH26.7	347	0	0	0	0	4	19	108	168	47	1	0	0	(13)	
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)	
(15)	Precipitation	PrecAvg	728	48	41	51	66	91	120	50	54	58	51	52	46	(15)	
(16)		PrecMax	1089	110	76	77	130	214	322	183	146	146	123	96	90	(16)	
(17)		PrecMin	520	14	10	17	28	13	21	5	9	6	13	17	23	(17)	
(18)		PrecStd	146	20	15	17	21	47	73	42	35	32	25	21	17	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.7	14.4	24.1	27.0	29.7	31.9	33.4	30.5	23.5	16.3	10.8	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	8.3	8.5	12.2	19.4	24.9	25.8	29.4	30.5	28.0	20.0	12.9	7.5	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	6.4	6.5	10.3	15.8	22.7	23.8	27.5	28.4	25.6	17.0	11.1	5.6	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	5.1	4.6	8.0	13.0	19.9	21.8	25.7	26.2	22.7	14.1	10.1	4.1	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range		MDBR	10.5	11.0	11.0	11.8	12.8	12.9	15.3	15.7	14.5	12.4	10.2	10.2	(35)	
(36)		5% DB	MCD BR	10.4	11.5	13.3	17.0	17.6	16.5	18.6	19.6	19.7	16.3	11.1	10.6	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCD BR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.247	0.261	0.284	0.339	0.345	0.338	0.340	0.357	0.320	0.284	0.260	0.233	(40)		
(41)		taud	2.400	2.395	2.421	2.301	2.369	2.436	2.437	2.377	2.457	2.537	2.489	2.417	(41)		
(42)		Ebn at Noon	835	906	937	907	913	920	912	879	883	869	815	807	(42)		
(43)		Edh at Noon	62	80	94	118	116	109	107	109	90	70	57	53	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.24	3.47	4.80	5.72	6.39	7.09	5.66	4.02	2.49	1.39	0.99	(44)		
(45)		RadStd	0.10	0.17	0.30	0.32	0.50	0.49	0.39	0.35	0.33	0.25	0.11	0.09	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)